

Towards Operational Resiliency with Real-Time Sensor Analytics using Machine Learning

Abstract—Medium and Heavy-Duty (MHD) vehicles are crucial in both civilian and military infrastructure for logistics and operations. The resilience of their embedded network systems, especially in hostile environments, is vital for maintaining operational continuity amidst cyber-attacks or kinetic malfunctions. This study focuses on enhancing the resilience of vehicle sensor systems within the SAE J1939 networking architecture. We propose a predictive algorithm, implemented on an embedded device, to maintain system functionality despite sensor disruptions. The algorithm uses alternative sensor inputs to predict missing sensor values, ensuring continuous operation of the vehicle. Our strategy entails training a specialized machine learning model to predict sensor values in the event of partial system failures. This empowers the machine learning output to seamlessly substitute for missing sensor data during instances of sensor disconnects. A proof of concept is demonstrated on a Class 6 heavy vehicle with a diesel engine, highlighting the model’s effectiveness in maintaining operational capacity under disrupted conditions. This research not only showcases the potential of predictive algorithms in mitigating sensor failures but also has broader implications for improving the resilience of cyber-physical systems in challenging operational environments.

Index Terms—SAE-J1939 network architecture, EGR - Exhaust Gas Recirculation, MitM - Man in the Middle, NTC - Negative Temperature Coefficient.

I. INTRODUCTION

As the US heavy-duty vehicle fleet surpasses 12 million units, their role in the national economy and infrastructure becomes increasingly significant [1]. However, this growing reliance on sophisticated electronics, especially the Controller Area Network (CAN) bus, introduces vulnerabilities to cyber-attacks and mechanical faults. Current Intrusion Detection Systems (IDS) offer limited protection by only identifying threats without addressing the critical need for system resilience.

This research explores the real-time integration of detection and machine learning-based correction within the CAN bus system, employing Long Short Term Memory (LSTM) auto-encoders as suggested by Shirazi et al. [2]. Our approach involves leveraging data from functional sensors to precisely estimate values for faulty ones, as depicted in the data flow diagram in Figure 1. Using an external computational node on the truck system, we interpret the CAN bus network, decoding and converting sensor values. The live data stream is analyzed for anomalies, triggering a machine learning model to predict missing sensor values from past session data. Predicted analog sensor values are sent to the Engine Control Module (ECM), while CAN-based values are transformed into CAN frames and reintegrated into the network. This dual strategy ensures operational safety, addressing challenges from cyber threats and sensor tampering.

Hence, our research question is: *How can we effectively integrate real-time detection with machine learning-based data correction within the CAN bus system of heavy-duty vehicles to enhance their cybersecurity and operational resilience?*

This study aims to not only identify cyber threats but also to actively counteract them, ensuring continuous and safe operation of heavy-duty vehicles amidst cyber-attacks or sensor failure.

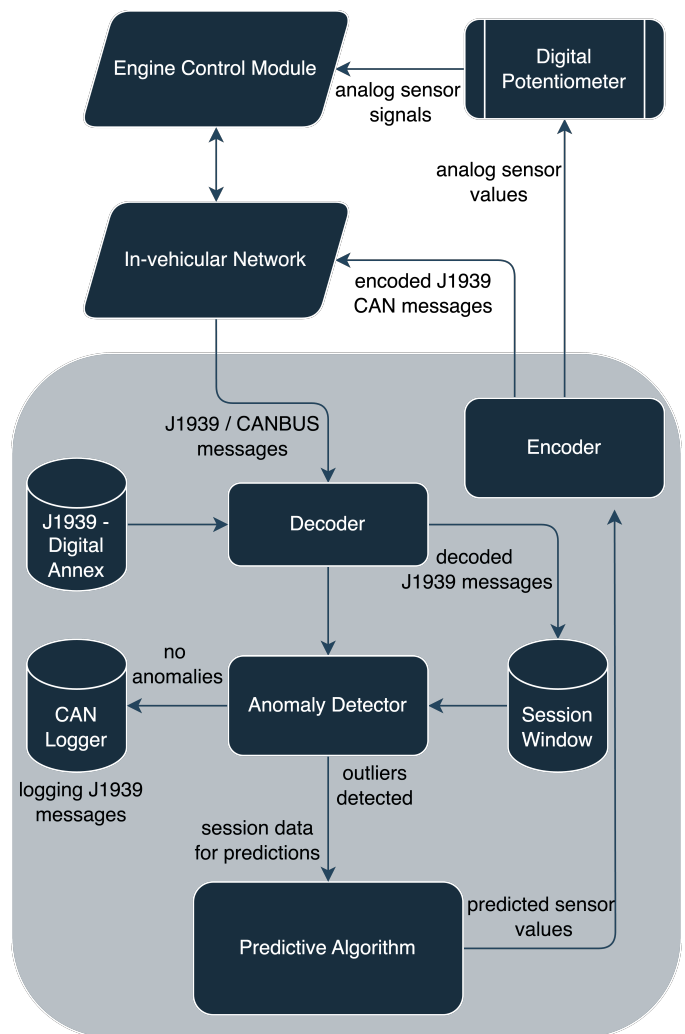


Fig. 1: Data-flow Diagram

The rest of the paper is organized as follows. In section II we discuss the background on heavy vehicle network, with emphasis on SAE J1939 protocol. In section III we

discuss the research in this field, limitations to existing work, and formulate our approach. In section IV we discuss the experiments performed so far, and in section V we discuss our plans on how we will conduct the remaining experiments.

II. BACKGROUND

Modern trucks are equipped with advanced sensors, electronic controls, and data communication systems, enabling real-time monitoring, diagnostics, and the integration of new capabilities through enhanced connectivity. A critical component in this evolution is the SAE J1939 communication protocol, which standardizes a CAN bus network for Electronic Control Units (ECU)s in heavy-duty vehicles. This protocol facilitates interoperability between various system components and vendors by defining standard network addressing, messaging mechanisms, and data exchange formats. The SAE J1939 standard covers aspects like signaling levels, wiring standards, connector pinouts, and prioritized messaging with a 29-bit identifier, which ensures that high-priority vehicle control data takes precedence over lower-priority communications. It also includes standardized Parameter Group Numbers (PGNs) and Suspect Parameter Numbers (SPNs) to identify critical vehicle metrics like engine temperature and oil pressure, enabling seamless vendor interoperability. The protocol's ability to multiplex functions such as engine, transmission, and safety systems onto a single CAN bus network has been foundational in advancing vehicle connectivity, automation, and emissions reduction. Adopted in 1994, SAE J1939 has been instrumental in shaping the landscape of vehicle data communication in heavy commercial vehicles for over two decades.

III. RELATED WORK

A. Vulnerabilities in Vehicular Systems

Recent research has uncovered certain vulnerabilities within the SAE J1939 protocol used in heavy vehicle communications. Studies [3], [4] have highlighted the risks of impersonation and denial-of-service attacks that can compromise the safety and efficiency of these vehicles. Further investigations have revealed additional attack vectors, including spoofing attacks [5] and MITM attacks [6], which exploit the lack of encryption and authentication in automotive communication protocols. These findings emphasize the imperative for implementing advanced security measures grounded in machine learning intelligence within contemporary vehicular systems.

B. Proposed Countermeasures and Challenges

A variety of countermeasures have been proposed to mitigate these vulnerabilities. Solutions involving cryptographic protections, anomaly detection systems, and securing ECUs have been explored [7], [8]. However, these approaches face implementation challenges, including computational overhead and compatibility issues, which limit their practical application in real-world scenarios.

C. Machine Learning in Intrusion Detection

Intrusion detection models utilizing Deep Neural Networks (DNNs) have been proposed [9] to distinguish between normal and abnormal packets in vehicular networks. These models analyze internal network traffic to generate real-time warning alarms with better precision and reliability. Additionally, machine learning models have been employed to reconstruct compromised sensor data [2], [10], using techniques like LSTM autoencoders. However, these methods typically require a separate neural network for each sensor, leading to substantial computational demands.

D. Our Approach

Our research introduces a novel, unified predictive model, deployed on a microcontroller, capable of reconstructing compromised sensor data across multiple sensors efficiently. This model addresses both the computational limitations and the necessity for robust cybersecurity.

Our revolutionary approach, depicted in Figure 1, marks a breakthrough in vehicular cybersecurity. Through innovative techniques, we enhance real-time data integrity within heavy vehicle networks. At its core, our methodology features a sophisticated decoder proficient in interpreting live CAN traffic, providing deep insights into ongoing data communication dynamics. This enables the system to predict and proactively address missing sensor data in real-time. The seamless integration of predictive capabilities not only anticipates but also promptly corrects information gaps, ensuring the reliability and security of the entire system. Simultaneously, an encoder program fortifies heavy vehicle networks against compromised data, significantly boosting the overall cybersecurity resilience of the vehicular ecosystem.

The decoder program, illustrated in Figure 2, plays a pivotal role in interpreting real-time CAN traffic within heavy vehicle networks, leveraging insights from the J1939 digital annex document. It decodes and archives values in a logger database for comprehensive drive analysis, concurrently managing a session database for specific instances. Anomalies, such as missing or compromised sensor values during a drive, trigger an advanced prediction algorithm. This algorithm utilizes stored session data to predict missing sensor values in real-time, enhancing the security of heavy vehicle networks.

Upon anomaly detection, the system seamlessly activates the robust prediction algorithm outlined in Figure 3. This time series forecasting model, pre-trained on high-performance computational devices using diverse datasets, is deployed on a microcontroller. The model processes live test data against learned patterns, ensuring adaptability to varied environmental conditions. This integration establishes a resilient vehicular cybersecurity framework.

In tandem with the prediction algorithm, our cybersecurity architecture features a crucial encoder program as shown in Figure 4. Tasked with encoding machine learning predictions into CAN frames, this program seamlessly reintegrates the CAN based sensor predictions back into the in-vehicle network. Also, the encoder program demonstrates versatility

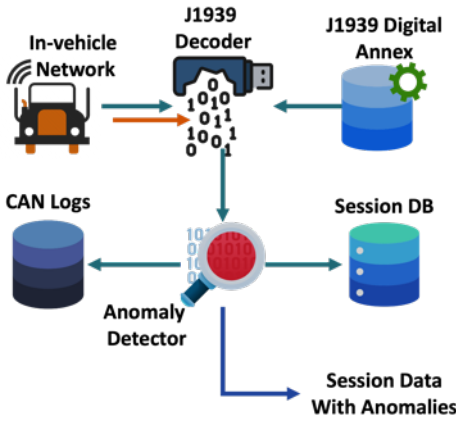


Fig. 2: Decoder Application

by adeptly handling analog sensor values, using a digital potentiometer for precise conversion into analog signals. This adaptation ensures seamless integration of machine learning insights. In cases of missing sensor values, encoded analog signals are sent directly to the ECM.

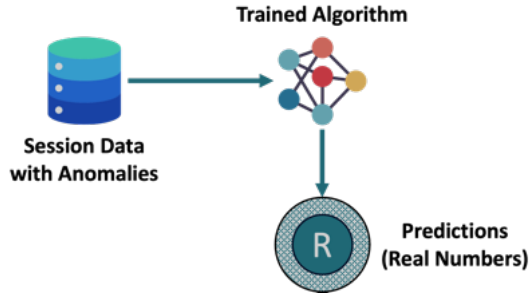


Fig. 3: Prediction Model

Effective utilization of data-driven modeling for predictive intelligence relies heavily on seamless integration into real truck systems. However, existing solutions face challenges related to scalability and computational efficiency, especially in resource-constrained environments. The requirement for individual models for each sensor exacerbates this burden. Therefore, alongside the development of prediction algorithms, it is imperative to assess physical integration and demonstrate hardware feasibility for transmitting forecasted parameters to relevant vehicle subsystems. Our approach addresses this issue by minimizing computational load through an external device, scaling across all vehicular network sensors using a unified machine learning model.

IV. SENSOR EXPERIMENTS

Our approach underscores the dual focus on advancing sophisticated prediction algorithms and seamlessly integrating these capabilities into authentic truck systems. Recognizing that the efficacy of data-driven modeling hinges on the harmonious fusion of existing components, we extend our emphasis

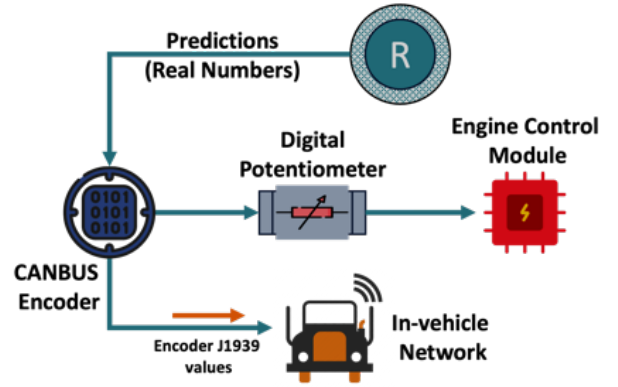


Fig. 4: Encoder Application

beyond algorithmic prowess to include tangible hardware integration for transmitting forecasted parameters into operational vehicle subsystems.

This segment outlines our experiments in hardware integration for predictive engine and transmission controllers in real trucks. Through the use of test benches and ECU emulators, we meticulously assess real-time data flow, communication interfaces, and integration capabilities. The results of these experiments showcase the system's adeptness in transmitting key forecasted parameters, forming the foundation for broader validation across commercial vehicle fleets. Leveraging a Raspberry Pi as a computational node connected to the vehicle's OBD II port, our experiments achieve a comprehensive assessment, ensuring end-to-end technical validation. This versatile setup affirms the seamless integration of our algorithms into practical truck systems within real-world operational environments, establishing a robust framework for the convergence of predictive intelligence and hardware integration.

A. Test Bed Setup

J1939 CAN traffic is generated using an engine control module as shown in Figure 5. A MITM device inserted into the network manipulates the traffic to simulate compromised or missing sensor data. A microcontroller (Raspberry Pi) connects to the J1939 network via the ECM's OBD-II port. It monitors the sensor data stream for anomalies. Whenever missing values are detected, the microcontroller triggers the predictive algorithm to estimate replacements. By leveraging controlled injection of sensor anomalies in the testbed, we can validate the system's ability to identify disturbances and reconstruct disrupted signals in real-time CAN bus environments. The modular setup with live sensor data generation, MITM attack simulation, and microcontroller prediction enables end-to-end testing of the robust anomaly response framework.

1) *MITM*: Conducting a MITM simulation assesses and fortifies communication security between sensors and the ECM in vehicles. Controlled scenarios mimic unauthorized data interception and manipulation, identifying potential vulnerabilities. Emulating realistic automotive networks with the CAN bus and various MITM attack scenarios, the real-time

Test No	Test case description / status of the sensors	Error Status	Check Engine Light	Diagnostics tool Response
1	Disconnect one CAN sensor from the engine	Engine ECU generates fault codes on the network	Yes	Fault code for that sensor is Displayed
2	Simulate one CAN sensor to the engine using Raspberry Pi	No Fault Codes	No	No Fault Code
3	Disconnect all CAN sensors from the engine	Engine ECU generates multiple fault codes on the network	Yes	Multiple fault codes are displayed for multiple sensors
4	Simulate all CAN sensors to the engine except VGT (Variable Geometry Turbocharger), using Raspberry Pi	No Fault Codes	No	No Fault Code
5	Simulate constant readings of all CAN sensors to the engine except VGT (Variable Geometry Turbocharger) using Raspberry Pi	No Fault Codes	No	No Fault Code

TABLE I: Tests validating hardware integration feasibility of CAN sensors

implementation on a research vehicle, facilitated by a micro-controller, provides practical insights into risks associated with compromised sensor-ECM communication. Outcomes will guide strategies for reinforcing security measures, ensuring the reliability of sensor data in modern vehicles.

In a series of experiments, two sensor types were scrutinized: those within a CAN architecture, communicating digitally, and analog sensors producing signals based on measured quantities. CAN-based sensors engage the ECM through electronic messaging, while analog sensors, like the EGR temperature sensor, employ NTC thermistors for temperature measurement. Analog signals are commonly in the form of changes in voltage, resistance, or current. This dual approach provides valuable insights into sensor performance under diverse engine conditions. Evaluating both digital and analog technologies enhances understanding of complex interactions within modern automotive engines, guiding future strategies for sensor integration, redundancy, and optimization to balance performance, cost, reliability, accuracy, and safety of critical systems.

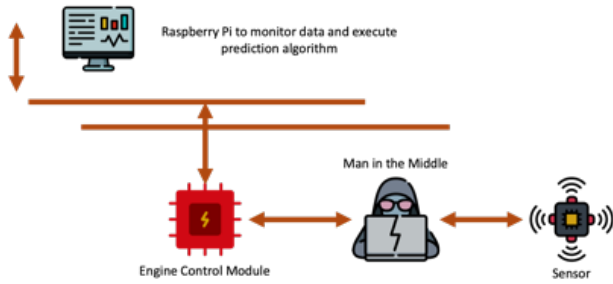


Fig. 5: Man in the Middle simulation

2) *CAN Sensor simulation*: This experiment focuses on evaluating the vulnerabilities introduced by a MITM attack on a CAN-based NOx sensor in a truck's exhaust system. The NOx sensor plays a crucial role in determining the concentration of NO and NO2 molecules, influencing the engine's air and fuel ratio for optimal combustion. The sensor's output is transmitted on the CAN bus and read by the ECU. The experiment involves strategically inserting a MITM device, such as a Teensy or Raspberry Pi, to manipulate the sensor readings. Initially, the NOx reading is within the expected

range at -151.15 ppm as illustrated in Fig. 6. However, the MITM attack successfully manipulates the value to an extreme 3,076.75 ppm as depicted in Fig. 7, exceeding the sensor's limits. This demonstrates the potential for compromised data links to inject incorrect sensor readings, leading to poor emissions control and reduced engine performance.

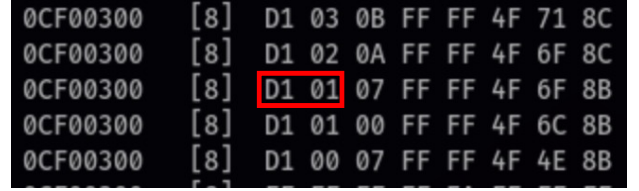


Fig. 6: Before data manipulation

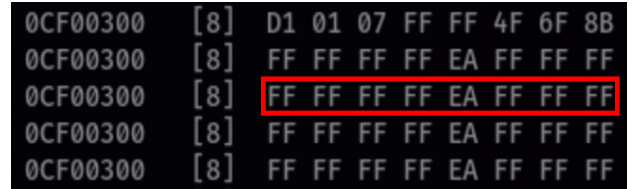


Fig. 7: After data manipulation

The experiment's significance lies in revealing the practicality of seamlessly integrating machine learning-predicted values into the ECU in the event of sensor failure. This cyber-physical approach offers a resilient solution for real-world applications by ensuring uninterrupted operation during faults. Exploratory test cases further investigate system resilience when CAN communication is intentionally compromised as exemplified in Table. I. Disconnecting CAN-based sensors triggers fault codes and illuminates the check engine light, prompting detailed diagnostic trouble codes. Similar experiments involve intentionally disconnecting CAN sensors and simulating output values. The results indicate that machine learning algorithms can fill gaps caused by sensor failures. In the event of a CAN sensor malfunction, predicted values from machine learning models can be encoded into CAN bus messages and reintegrated into the network, providing sensor redundancy. Ongoing testing aims to validate reliability im-

provements against diverse failure scenarios before integrating this cyber-physical technique into the vehicle's system.

3) *Analog sensor simulation*: This experiment focuses on enhancing the reliability of analog sensors crucial for monitoring temperature and pressure within a vehicle's engine components. Specifically centered on the engine of the vehicle, the analog sensors exhibit changes in resistance and voltage, corresponding to fluctuations in temperature. The experiment takes a forward-looking approach by deploying a machine learning algorithm to forecast temperature and pressure, surpassing mere predictive capabilities. The breakthrough lies in the integration of these predictions through a mapping process onto equivalent resistance values using a digital potentiometer.

The simulated analog sensor methodology demonstrates adaptability and practicality in transmitting equivalent resistance values to the ECU. The spotlight of this experiment is on the EGR temperature sensor, leveraging Negative Temperature Coefficient (NTC) thermistors. Previous experiments paved the way by substituting failed digital CAN bus sensor data with machine learning model outputs in CAN messages. Analog sensor failures were similarly addressed through simulated fault injection and predictive analytics. The experiment culminates in a cyber-physical loop, where machine learning-predicted EGR temperatures seamlessly substitute for missing analog data during sensor disconnects. The correlation between sensor resistance adjustments and simulated temperature readings, detailed in Table II, solidifies the real-world applicability of this approach. Equations 1 and 2 provide a mathematical foundation modeled by Steinhart and Hart, 1968 [11], establishing a clear relationship between resistance and temperature for NTC thermistors in this groundbreaking experiment.

Sample No.	R (kΩ)	V (V) across resistance	T (°C)
1	100	2.469	41.1
2	200	3.288	22.8
3	300	3.698	9.9
4	400	3.943	2.2
5	500	4.108	-5.4
6	600	4.225	-13
7	700	4.313	-18.3
8	800	4.381	-21.6
9	900	4.435	-23.7
10	1000	4.48	-25.2

TABLE II: EGR Temperature sample readings

The intricate relationship between resistance and temperature, as captured by Equation 1, empowers the anticipation of sensor resistance for any given temperature. Fig. 4 visualizes this process, showcasing how machine learning-predicted temperatures seamlessly transition into corresponding sensor resistance values. The final step involves the conversion of these resistance values into analog voltage signals, facilitated by a digital potentiometer.

$$RT = \exp \left(\left(\frac{\left(\frac{B}{3C} \right)^3 + \left(\frac{A - \frac{1}{T+273.15}}{C} \right)^2}{4} \right)^{\frac{1}{2}} - \frac{\left(\frac{A - \frac{1}{T+273.15}}{C} \right)^{\frac{1}{3}}}{2} \right) - \left(\left(\frac{\left(\frac{B}{3C} \right)^3 + \left(\frac{A - \frac{1}{T+273.15}}{C} \right)^2}{4} \right)^{\frac{1}{2}} + \frac{\left(\frac{A - \frac{1}{T+273.15}}{C} \right)^{\frac{1}{3}}}{2} \right) \quad (1)$$

Where 'T' is 'Predicted Temperature' and 'RT' is 'Resistance at Temperature', and A, B, C are the coefficients:

$$A = 0.000528023348$$

$$B = 0.000225611448$$

$$C = 5.60 \times 10^{-8}$$

$$T = \frac{1}{A + B \cdot \ln(RT) + C \cdot (\ln(RT))^3} - 273.15 \quad (2)$$

The thermistor's resistance, influenced by temperature, is accurately described by the Steinhart-Hart model [11], enabling precise calculations using equations 1 and 2. However, this can strain computational resources, especially in embedded systems. A more efficient alternative involves creating lookup tables from experimental data, mapping temperatures to corresponding resistances. This approach minimizes computational load, enhancing the operational efficiency of embedded systems. Employing lookup tables based on experimental measurements offers a simplified and effective method for determining resistance from temperature in practical device implementations.

By successfully encoding digital predictions into analog signals, this experiment serves as a pivotal step in the cyber-physical integration paradigm. This approach not only mimics the real transducer output but also establishes a resilient bridge between the digital and physical realms. Consequently, our demonstrated methodology stands as a tangible and effective contribution to the advancement of system resilience, paving the way for practical implementations in scenarios where virtual sensor measurements can seamlessly substitute for faulty physical sensors.

B. Machine Learning Experiments

Having demonstrated the capability of transforming machine learning predictions into sensor-representative analog signals, the next consideration is the feasibility of deploying such data-driven models on embedded automotive hardware. Recognizing the computational constraints of microcontroller

devices, a pre-trained machine learning model is employed to predict missing sensor values in real-time. Initially trained on a high-performance system, a streamlined version, focusing solely on inference, is extracted for deployment. Computational performance experiments reveal that generating 2,000 sensor value predictions on a Raspberry Pi takes approximately 4 seconds using an LSTM model. With the simplified predictions-only model, optimized software, and hardware acceleration, the processing time is anticipated to meet real-time requirements, aligning with the space and time complexity constraints of an in-vehicle embedded device.

In order to facilitate real-time implementation, we employ a strategy involving the creation of multiple models, each trained on diverse test datasets to ensure accurate predictions. Vehicle data is collected under various drive conditions, considering the potential deviation of incoming data from the training dataset. Each model undergoes training using a single cycle of dataset incorporation from a collection of different dataset cycles. Following this, the generated models undergo testing using distinct dataset cycles that differ from the training dataset. The resulting error percentages for each tested dataset with every model are then plotted and compared. This comprehensive analysis is crucial for evaluating the accuracy and performance of the models when confronted with datasets different from those used in their initial training.

Ensuring consistent accuracy in real-time predictions for vehicle sensor values is paramount for our project. The dynamic nature of sensor data during diverse driving conditions demands a robust machine learning model. To guarantee reliability, we meticulously map datasets collected under various conditions to corresponding trained models. Each drive cycle presents unique sensor values, emphasizing the need to maintain the model's stability across diverse scenarios. Our approach involves leveraging a comprehensive dataset that covers a spectrum of driving conditions, enabling the machine learning model to generalize effectively. By addressing the inherent variability in real-world driving, our objective is to establish a resilient and dependable model capable of delivering consistent and accurate predictions in real-time environments, thereby enhancing overall vehicle performance and safety.

DataCycle	Dataset	Trained Model
Engine Idling	Dataset A [DA]	Model A [MA]
City Drive	Dataset B [DB]	Model B [MB]
Highway Drive	Dataset C [DC]	Model C [MC]
Long Drive	Dataset D [DD]	Model D [MD]

TABLE III: Dataset and Model generation

In Figure 8, a dataset is gathered from a research truck under various driving conditions, as outlined in Table III. Subsequently, a machine learning model is crafted using this dataset as the training data, as depicted in Figure 8. To evaluate the model's robustness, testing is executed with distinct datasets (B, C, D) not utilized during the training of a specific model (e.g., Model A trained solely on dataset A). Sixteen experiments are performed, each involving a unique

combination of training and testing datasets. The resultant error percentages are meticulously documented for comparative analysis among all generated models. This thorough assessment aims to identify a generalized model exhibiting optimal accuracy and minimal error, providing valuable insights for selecting the most effective model. The model demonstrating the highest accuracy and lowest error percentage is highlighted as the superior choice based on the experimental outcomes.

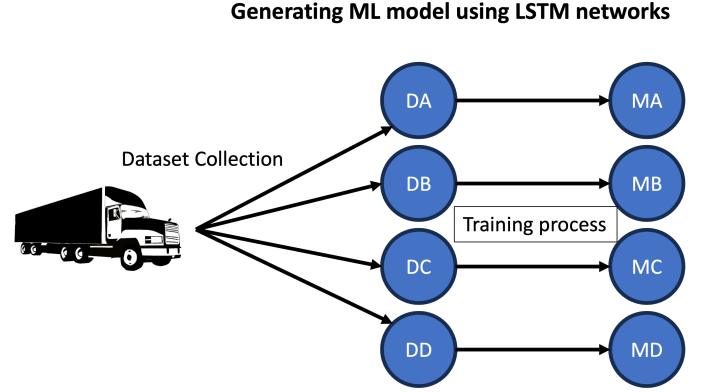


Fig. 8: ML model testing process

Among the conducted experiments, Model B emerges as the most effective, exhibiting superior performance with significantly lower error percentages in the identification of missing SPNs compared to other models. The outcomes are graphically depicted in Figure 9, 10 where the error percentages for Model B are plotted across various datasets, namely [A, B, C, D]. This graphical representation provides a concise visualization of Model B's robustness and accuracy when subjected to diverse testing conditions. The observed trend underscores the model's consistent reliability in detecting missing SPNs across a range of datasets, reinforcing its prominence as a standout choice among the evaluated models.

Model B, derived from city drive data, undergoes testing with dataset D, obtained from a long drive spanning Fort Collins to Michigan. This unique testing combination yields exceptionally favorable outcomes. Notably, in the evaluation of 53 Signal Phase and Timing (SPN) instances, 40 exhibit error percentages below 20%, while 6 SPNs achieve error percentages below an impressive 10%. These results, observed across various experiments, underscore the model's exceptional performance and highlight its proficiency in accurately predicting SPN values, particularly in extended drive scenarios.

In Figure 9 and 10, a bar graph depicts error percentages, employing a spectrum of colors for distinction. Lighter shades of blue denote lower error percentages, while darker hues indicate higher error percentages. This visual representation allows for an intuitive interpretation of the varying degrees of accuracy across the depicted data, with the color gradient serving as a key indicator of the magnitude of errors.

Our research is aimed on enhancing the effectiveness of the algorithms utilized. We conducted thorough evaluations by training these algorithms across a spectrum of data scenarios,

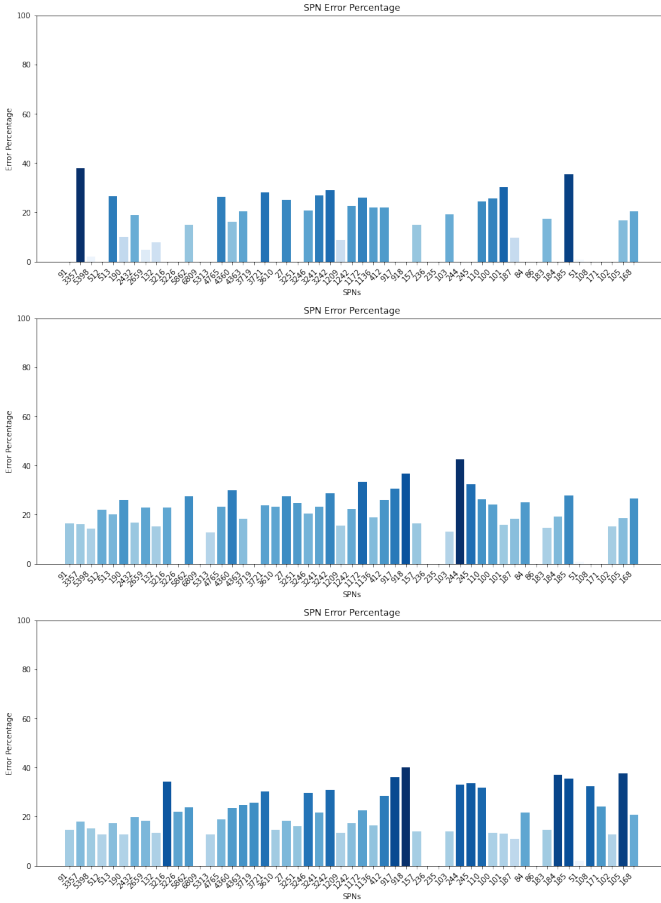


Fig. 9: ML model B results with dataset [A, B, C]

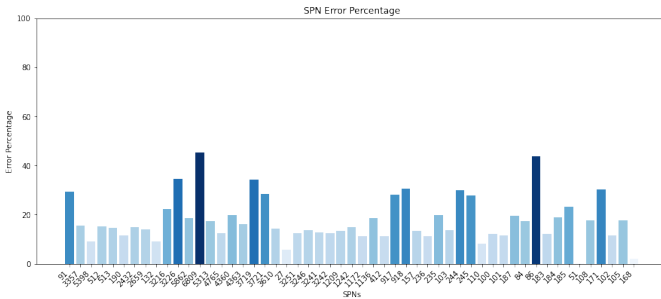


Fig. 10: ML model B results with dataset D

particularly emphasizing the examination of missing values—a prevalent challenge encountered in real-world contexts. To gauge the accuracy of our results during testing, we utilized the Average Percentage Error (err) as our metric.

$$err = \frac{1}{n} \sum_{i=1}^n \frac{|e_i|}{z} \times 100 \quad (3)$$

This metric is calculated by computing the absolute error between the actual and predicted values for each data point, denoted by $|e_i|$. It is then normalized against the range of

the specific SPN, represented by z , where n encapsulates all dataset points for the specific SPN.

C. Additional Experiments

In this research endeavor, we aim to explore and evaluate the efficacy of various deep learning algorithms, including sparse binary and dense binary models, through additional experiments. As depicted in Figure 11, our investigation entails training these algorithms across a spectrum of recurrent neural network (RNN) and deep learning architectures. By conducting these experiments, we seek to gain deeper insights into the comparative performance, strengths, and weaknesses of these diverse algorithms in tackling complex tasks. Such an empirical analysis holds significant promise in advancing our understanding of the nuances inherent in different deep learning approaches, thereby paving the way for more informed algorithmic selections and optimizations in real-world applications.

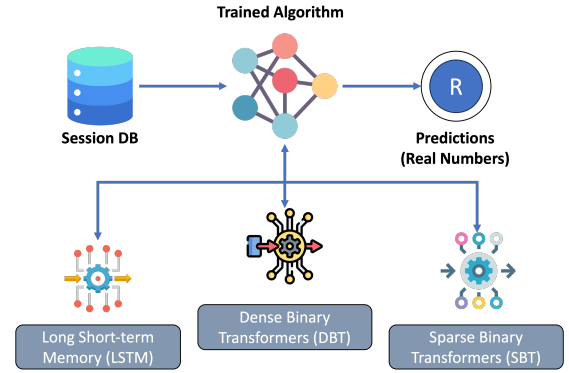


Fig. 11: Various Training algorithms

V. FUTURE WORK

Our future work will focus on refining sensor value reconstruction in heavy-duty vehicles to counteract both kinetic and cyber threats. This includes expanding experiments on CAN and analog sensor signals and enhancing the predictive accuracy of our machine learning models for real-time applications. We also aim to develop comprehensive mappings to better understand sensor relationships and integrate these findings into vehicle systems. Additionally, we plan to explore driving pattern recognition to identify potential vulnerabilities and validate the practicality of machine learning predictions in real-world scenarios. This research is geared towards creating a robust, intelligent system that ensures safe and efficient operation of critical vehicle components, blending advanced sensing technologies with data-driven insights. Our future work will focus on several key areas:

- 1) **Testing on different ML models:** We will continue to test the prediction accuracy on various other machine learning models using:
 - LSTM [current work in progress]
 - Dense binary transformers

- Sparse binary transformers

perform the the existing test cases with all these models which would result in 36-48 different experimental results.

- 2) **Optimizing Pretrained Models:** We will continue refining our pretrained models to ensure faster and more accurate real-time predictions, particularly focusing on reducing response times and enhancing prediction accuracy.
- 3) **System Integration and Validation:** Emphasis will be placed on the seamless reintegration of predicted values into the in-vehicle network, with rigorous validation processes to ensure system adaptability and reliability.
- 4) **Exploring Driving Pattern Recognition:** To further enhance vehicle resilience, we plan to develop models capable of understanding and predicting driving patterns, thereby mitigating potential vulnerabilities.
- 5) **Validating Machine Learning Predictions:** A significant part of our future work will involve validating the feasibility and reliability of machine learning predictions in real-world vehicle systems.
- 6) **Addressing Security Concerns:** Recognizing the importance of cybersecurity, we will explore the uncharted realms of attack surfaces to fortify the security of our systems against emerging threats.
- 7) **Implementing Raspberry Pi:** The ultimate deployment plan involves integrating a lightweight model onto a microcontroller device, specifically a Raspberry Pi. This model will seamlessly become part of the real-time heavy vehicle system, enabling on-the-fly predictions for missing sensor values. This strategic integration onto the Raspberry Pi aligns with the goal of optimizing computational resources while enhancing the system's capacity to deliver accurate real-time predictions within the heavy vehicle environment in real-time.

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